Mathematical Formulation of HydroBoost – a Hydro-Hybrids Valuation Tool

Model C – Pumped Storage Hydro (PSH) Plant

March-2026

1. Summary

This document presents the mathematical formulation of the HydroBoost model, a weekly profit-maximization tool developed to support short-term investment decision-making for hybrid hydropower systems. It is expected to serve as the central optimization module for Subtask 2.3 of the project “Hydro+Storage: Accelerating Industry Deployment of Hydropower Hybrids”.

The HydroBoost model is developed to simulate the short-term operation of a grid-connected, co-located hybrid energy system consisting of three main components: a pumped storage hydropower (PSH) system, a battery energy storage system (BESS), and non-dispatchable renewable energy sources (RES), such as wind and solar. In this setup, RES generators are modeled as unidirectional, they can only deliver energy, either directly to the grid or to storage systems. In contrast, BESS and PSH systems are bidirectional, capable of both injecting and withdrawing power depending on their operating mode (discharging vs charging, generation vs pumping). The model supports co-optimization across both energy and ancillary service markets, including regulation-up, regulation-down, and spinning reserve. The level of ancillary service participation depends on the specific hydro configuration adopted (e.g., generation vs pumping mode). The objective function is to maximize the net market revenue over a weekly planning horizon, by jointly optimizing energy dispatch, storage operation, and ancillary service provision on an hourly basis.

The research is being conducted by a team of experts from Argonne National Laboratory (ANL) and Idaho National Laboratory (INL), and National Renewable Energy Laboratory (NREL). The objective is to provide an investment decision support tool that can demonstrate the technical and economic advantages of operating a hydroelectric plant in conjunction with a BESS.

2. Summary of model structure, assumptions, and variants

2.1 Model structure and assumptions

- A single hydroelectric plant with one or more hydro generators and pumps (turbine-generator and pump-motor sets)
- Neglecting net head effect of stored water over performance curve of hydro generators
- Neglecting effect of hydraulic losses
- Hydro power generation and hydraulic short-circuit (HSC) operations represented as non-convex piecewise linear function
- Rough zones represented by complementarity constraints

- Using concept of regulation signal to obtain quantitative estimation of net energy delivered by BESS in real time due to deployment of regulation up and down services [1,2]
- Problem recast as Mixed Integer Linear Programming (MILP) model
- BESS model supported by [1,2,3,4] and hydropower model by [5,6,7].
- Ancillary services provided by both BESS and hydroelectric plant.
- Incorporation of pumped storage hydropower with hydraulic short circuit (PSH-HSC) representation, following the modeling framework described in [7].
- Incorporation of non-dispatchable renewable resources.

2.2 Model Variants

To ensure flexibility and broad applicability, three model variants are considered (see Fig.1A, 1B, and 1C).

Model A - Hydroelectric System without Reservoir

This variant retains the inflow term in the water balance equation for consistency but does not track water storage over time (i.e., there are no constraints on reservoir volume bounds or terminal storage). It represents a run-of-river system with no storage capability and limited flexibility, which is why no ancillary services are modeled for this configuration.

Model B - Hydroelectric System with Reservoir (Cascading Modeling)

This configuration includes natural inflow, dynamic water storage tracking, and spillage constraints. It allows for the provision of ancillary services (regulation up, regulation down, and spinning reserve) exclusively through the hydro generators. It models full reservoir dynamics with volume limits and initial/final storage constraints. In cascading modeling, multiple hydro plants are hydraulically linked so upstream releases and spillage become downstream inflows, coupling dispatch and reserve decisions across plants and time.

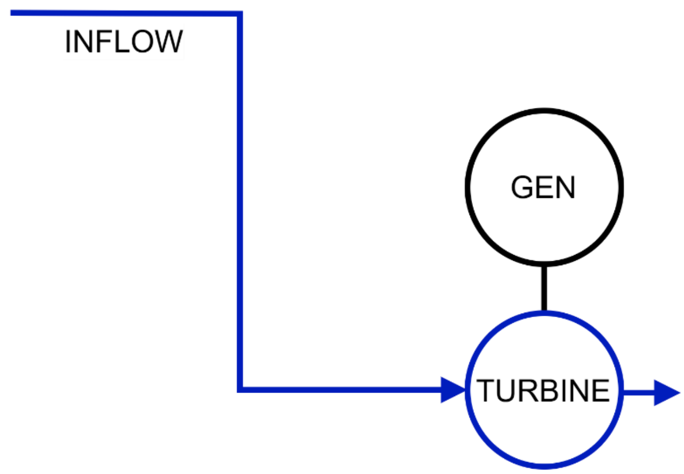
Model C - Pumped Storage Hydro (PSH) Plant

This model features a closed-loop configuration, excluding natural inflows. It includes non-dispatchable renewable energy source (RES) units and supports both generating and pumping modes. All three ancillary services (regulation up, regulation down, and spinning reserve) are provided by the hydro generators during generation mode.

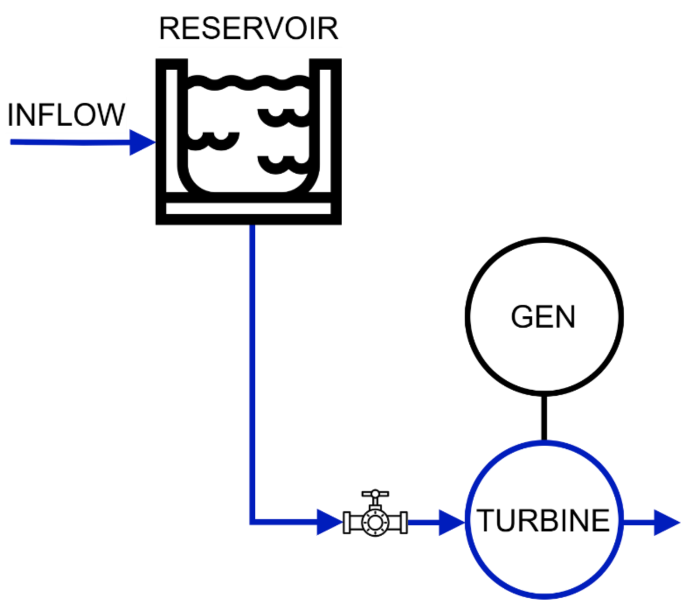
Model D - Pumped Storage Hydro (PSH) System with Hydraulic Short-Circuit (HSC) Operation

This model also uses a closed-loop configuration, excluding natural inflows. It includes non-dispatchable RES units and incorporates generating, pumping, and HSC modes. All three ancillary services (regulation up, regulation down, and spinning reserve) are provided by the hydro generators during both generation and HSC modes.

(A)



(B)



(C-D)

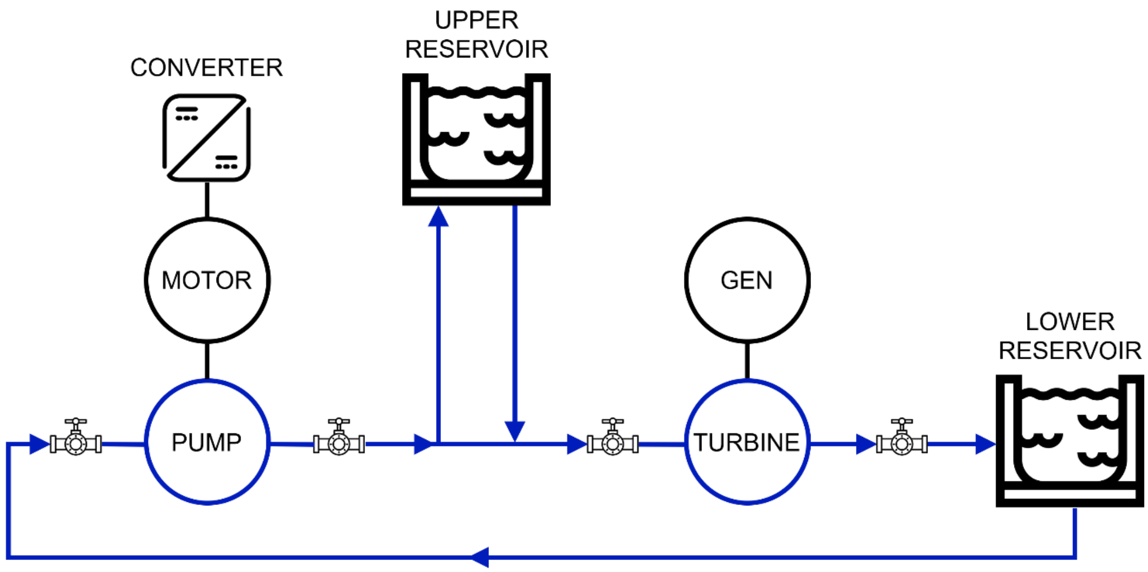


Fig. 1. Schematic representation of the three investigated configurations of the hydroelectric system: (A) Hydroelectric System with Reservoir, (B) Hydroelectric System without Reservoir, (C-D) Pumped Storage Hydro (PSH) System.

A complete schematic of the system layout considering Model C and Model D with the correspondent power flow pathways is shown in Fig. 2. Additionally, Fig. 3 presents the corresponding input/output power curve for HSC operation.

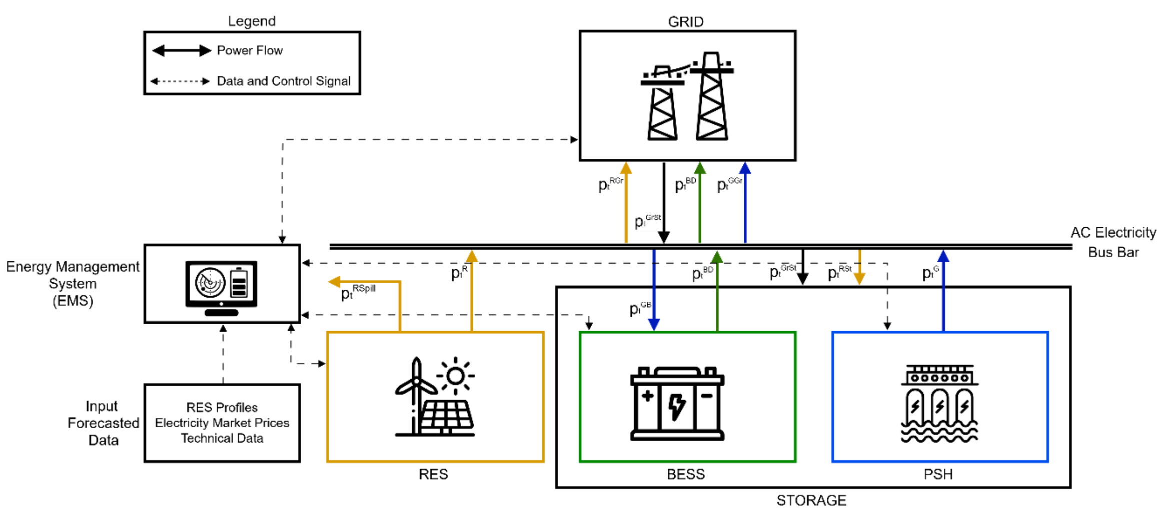


Fig. 2. Schematic representation of the grid-connected co-located hybrid asset under investigation in Model D, consisting of a pumped storage hydro (PSH) system, a battery energy storage system (BESS), and non-dispatchable renewable energy sources (RES).

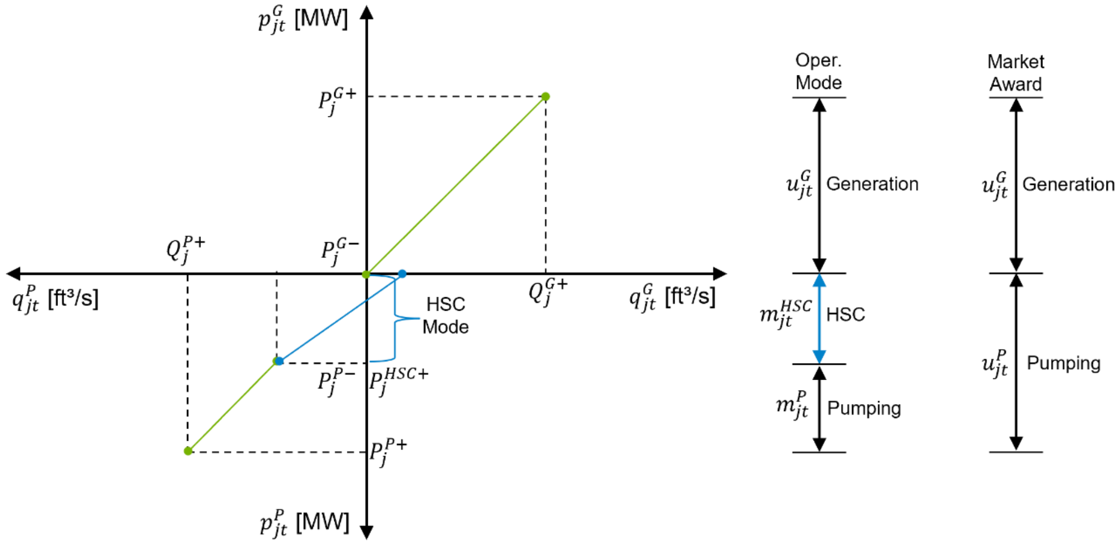


Fig. 3. Input/output curve for the PSH system with HSC mode. HydroBoost model uses binary variables to represent the multiple operating modes.

Nomenclature

Sets

$\mathcal{L}_j$	Set of linear segments in the performance curve of hydro generator $j \in \mathcal{U}$
$\mathcal{R}$	Set of non-dispatchable renewable power generators
$\mathcal{S}$	Set of storage units
$\mathcal{T}$	Set of periods, where $\mathcal{T} = \{1,2, \dots, N_{\mathrm{H}}\}$
$\mathcal{U}$	Set of hydro generators and pumps
$\mathcal{Z}_j$	Subset of blocks or segments of hydro generator j identified as rough zones of the unit, where $\mathcal{Z}_j \subseteq \mathcal{L}_j$

Indices

i	BESS unit index ($i \in \mathcal{S}$)
j	Hydro generator index ($j \in \mathcal{U}$)
k	Non-dispatchable renewable power generator ($k \in \mathcal{R}$)
t	Time segment ($t \in \mathcal{T}$)

Parameters

AET_i	Annual energy throughput limit for BESS unit i [MWh/yr]
CF	Factor to convert water flow units in $[\mathrm{ft}^3/s]$ into net volume units in [A-F/s]
C^{rz}	Penalization for operating within rough zone if honoring constraint is infeasible [$/(ft^3/s)]
C_0^{filt}	Coefficient to define the linear water filtration function [ft^3/s]
C_1^{filt}	Coefficient to define the linear water filtration function [$1/s$]
L_i^{cal}	Calendar lifetime of BESS unit i [yr]
L_i^{cyc}	Cycle lifetime of BESS unit i $[cyc]$
M	A large enough scalar needed for representation of rough zone constraints
N_D	Number of days in the look-ahead horizon
N_H	Number of hours in the look-ahead horizon
N_G	Number of hydro generators of the hydroelectric plant
N_P	Number of hydro pumps of the hydroelectric plant
P_i^{BC+}	Maximum charge capacity of BESS unit i [MW]

P_i^{BD+}	Maximum discharge capacity of BESS unit i [MW]
P_j^{G-}	Minimum power output of hydro generator j [MW]
P_j^{G+}	Maximum power output (or nominal capacity) of hydro generator j [MW]
P_j^{P-}	Minimum power consumption of hydro pump j [MW]
P_j^{P+}	Maximum power consumption of hydro pump j $[MW]$
P_{kt}^{RFor}	Forecast non-dispatchable generation of resource k in hour t [MWh]
Q_j^{G+}	Maximum water discharge of hydro generator j $[\text{ft}^3/s]$
Q_j^{G-}	Minimum water discharge of hydro generator j $[\text{ft}^3/s]$
$Q_j^{G\ell+}$	Maximum water discharge of block ℓ of performance curve of hydro generator
Q_j^{P+}	Maximum pumping level of hydro pump j $[\text{ft}^3/s]$
$R_j^{SU,G}, R_j^{SD,G}$	Startup ramp and Shutdown ramp of hydro generator j [MW/h]
$R_j^{U,G}, R_j^{D,G}$	Ramp up and ramp down of hydro generator j [MW/h]
RD_i^{B+}	Maximum Regulation-Down that can be allocated to BESS unit i [MW]
RU_i^{B+}	Maximum Regulation-Up that can be allocated to BESS unit i [MW]
RD_j^{G+}	Maximum Regulation-Down that can be allocated to hydro generator j [MW]
RU_j^{G+}	Maximum Regulation-Up that can be allocated to hydro generator j [MW]
SOC_i^-	Minimum state-of-charge of BESS unit i [MWh]
SOC_i^+	Maximum state-of-charge of BESS unit i [MWh]
SOC^{ini}	State-of-charge of BESS at beginning of period 1 [MWh]
SR_i^{B+}	Maximum Spinning Reserve that can be allocated to BESS unit i [MW]
SR_j^{G+}	Maximum Spinning Reserve that can be allocated to hydro generator j [MW]
SU_j^G, SD_j^G	Startup and shutdown cost of hydro generator j $[\$/\text{h}]$
SU_j^P, SD_j^P	Startup and shutdown cost of hydro pump j $[\$/\text{h}]$
$SU_j^{G,day}, SD_j^{G,day}$	Max number of startups and shutdowns per day of hydro generator j $[-]$
$SU_j^{P,day}, SD_j^{P,day}$	Max number of startups and shutdowns per day of hydro pump j $[-]$
T^{exp+}	Maximum capacity for transferring power to the grid for energy export [MW]
T^{imp+}	Maximum capacity for transferring power from the grid for energy import [MW]
V^-	Minimum water storage level of reservoir [A-F]
V^+	Maximum water storage level of reservoir [A-F]

ε	A very small number, such as $\varepsilon = 0.0001$, incorporated into the objective function to avoid simultaneous energy import and export
η_i	Roundtrip efficiency of BESS unit i [p.u]
η_i^C	Charging efficiency of BESS unit i [p.u]
η_i^D	Discharging efficiency of BESS unit i [p.u]
Θ_t^-	Total released water requirement in hour t $[\mathrm{ft}^3/\mathrm{s}]$
λ_t^E	Forecast day-ahead market price of energy in hour t, at the bus or area where the hybrid asset is located $[\$/\mathrm{MWh}]$
$\lambda_t^{E,RT}$	Forecast real-time market price of energy in hour t, at the bus or area where the hybrid asset is located $[\$/\mathrm{MWh}]$
$\lambda_j^{O\&M}$	Variable operation and maintenance O&M cost of hydro generator j [$/MWh]
$\lambda_t^{RU}, \lambda_t^{RD}$	Forecast day-ahead market price of regulation service (up and down) in hour t, at the bus or area where the hybrid asset is located $[\$/\mathrm{MWh}]$
λ_t^{SR}	Forecast day-ahead market price of spinning reserve SR in hour t, at the bus or area where the hybrid asset is located $[\$/\mathrm{MWh}]$
$\delta_t^{RU}, \delta_t^{RD}$	Fraction of regulation up (down) reserves sold in the day-ahead market, which are dispatched in real-time operation, where $\delta_t^{\mathrm{RU}}, \delta_t^{\mathrm{RD}} \in [0,1]$
$\hat{\rho}_j^G$	Average water-power conversion coefficient of hydro generator j $[\mathrm{MW}/(\mathrm{ft}^3/\mathrm{s})]$
$\rho_j^{G\ell}$	Slope of block ℓ of performance curve of hydro generator j $[\mathrm{MW}/(\mathrm{ft}^3/\mathrm{s})]$
ρ_j^P	Power-to-flow conversion coefficient for hydro pump j $[\mathrm{MW}/(\mathrm{ft}^3/\mathrm{s})]$
φ^{ini}	Scalar between 0 and 1 used to establish the volume of reservoir in the first day of the simulation [p.u.] (0 → reservoir is full - 1 → reservoir is empty)
φ^{end}	Scalar between 0 and 1 used to establish the volume of reservoir in the last day of the simulation [p.u.] (0 → reservoir is full - 1 → reservoir is empty)

Binary Variables

a_{jt}^G	Binary variable which is equal to 1 if hydro generator j is started at beginning of hour t, and 0 otherwise
a_{jt}^P	Binary variable which is equal to 1 if hydro pump j is started at beginning of hour t, and 0 otherwise
u_{it}^B	Binary variable which is equal to 1 when BESS unit i is set to charging mode, and 0 otherwise
u_{jt}^G	Binary variable which is equal to 1 if hydro generator j is on-line in hour t, and 0 otherwise
u_{jt}^P	Binary variable which is equal to 1 if hydro pump j is on-line at hour t, and 0 otherwise

y_t^P	Binary variable which is equal to 1 if hydro units are in pumping mode at hour t, and 0 if hydro units are in generation mode at hour t
z_{jt}^G	Binary variable which is equal to 1 if hydro generator j is shut down at beginning of hour t, and 0 otherwise
z_{jt}^P	Binary variable which is equal to 1 if hydro pump j is shut down at beginning of hour t, and 0 otherwise
$\psi_{jt}^{G\ell}$	Auxiliary binary variable required for representation of rough zone constraints in generation mode
$\omega_{jt}^{G\ell}$	Binary variable which is equal to 1 if water discharged by hydro generator j has exceeded block ℓ in hour t

Continuous Variables

e_{it}^B	State of charge of BESS unit i in hour t [MWh]
e_t^H	Water volume of reservoir in hour t [A-F]
p_{it}^{BC}	Power contributing to charge BESS unit i in hour t before accounting for losses [MWh]
p_t^{BC}	Energy contributing to charge all BESS units in hour t before accounting for losses [MWh]
p_{it}^{BD}	Power discharged from BESS unit i in hour t before accounting for losses [MWh]
p_t^{BD}	Energy discharged from all BESS units and accounted for at point of delivery in hour t [MWh]
p_{jt}^G	Power output of hydro generator j in hour t $[\mathrm{MWh}]$
p_t^{GrSt}	Power from grid allocated to storage (BESS and/or PSH) in hour t [MWh]
p_t^G	Power output of all hydro generators in hour t $[\mathrm{MWh}]$
p_t^{GB}	Hydro power allocated to charge the BESS in hour t $[\mathrm{MWh}]$
p_t^{GGr}	Dispatchable hydro power sold to the grid in the day-ahead market in hour t [MWh]
p_{kt}^R	Power output of non-dispatchable generator k in hour t [MWh]
p_t^{RGr}	Power of all non−dispatchable generators sent to grid in hour t [MWh]
p_{kt}^{RSpill}	Curtailed power of non-dispatchable generator k in hour t [MWh]
p_t^{RSt}	Power of non−dispatchable generators allocated to storage (BESS and/or PSH) in hour t [MWh]
p_t^R	Power of all non-dispatchable generators in hour t [MWh]
p_{jt}^P	Power consumption of hydro pump j in hour t [MWh]
p_t^P	Power consumption of all hydro pumps in hour t $[\mathrm{MWh}]$

q_{jt}^{G} Water discharge of hydro generator j in hour t [ft^3/s]

$q_{jt}^{G\ell}$ Water discharge of block ℓ of the piecewise linear function of hydro generator j in hour t when in generation mode [ft^3/s]

q_{jt}^{P} Water flow rate pumped through hydro pump j in pumping mode at hour t [ft^3/s]

q_{jt}^{P-TOT} Total water flow rate pumped through hydro pump j in pumping mode at hour t [ft^3/s]

$q_{jt}^{RU,G}$ Flow equivalent of regulation up allocated to hydro generator j in hour t, when in generation mode [ft^3/s]

$q_{jt}^{RD,G}$ Flow equivalent of regulation down allocated to hydro generator j in hour t, when in generation mode [ft^3/s]

$q_{jt}^{SR,G}$ Flow equivalent of spinning reserve allocated to hydro generator j in hour t, when in generation mode [ft^3/s]

r_{t}^{RU} Hydro + BESS reserve sold to market for regulation up in hour t [MWh]

r_{t}^{RD} Hydro + BESS reserve sold to market for regulation down in hour t [MWh]

r_{t}^{SR} Hydro + BESS reserve sold to market for spinning reserve in hour t [MWh]

$r_{it}^{RU,BC}$ Reserve for regulation up in charging mode provided by BESS unit i in hour t [MWh]

$r_{it}^{RD,BC}$ Reserve for regulation down in charging mode provided by BESS unit i in hour t [MWh]

$r_{it}^{SR,BC}$ Reserve for spinning reserve in charging mode provided by BESS unit i in hour t [MWh]

$r_{it}^{RU,BD}$ Reserve for regulation up in discharging mode provided by BESS unit i in hour t [MWh]

$r_{it}^{RD,BD}$ Reserve for regulation down in discharging mode provided by BESS unit i in hour t [MWh]

$r_{it}^{SR,BD}$ Reserve for spinning reserve in discharging mode provided by BESS unit i in hour t [MWh]

$r_{it}^{RU,B}$ Reserve for regulation up provided by BESS unit i in hour t [MWh]

$r_{it}^{RD,B}$ Reserve for regulation down provided by BESS unit i in hour t [MWh]

$r_{it}^{SR,B}$ Reserve for spinning reserve provided by BESS unit i in hour t [MWh]

$r_{t}^{RU,B}$ Reserve for regulation up provided by all BESS units in hour t [MWh]

$r_{t}^{RD,B}$ Reserve for regulation down provided by all BESS units in hour t [MWh]

$r_{t}^{SR,B}$ Reserve for spinning reserve provided by all BESS units in hour t [MWh]

$r_{jt}^{RU,G}$ Hydro reserve sold to market for regulation up provided by hydro generator j in hour t, when in generation mode [MWh]

$r_{jt}^{RD,G}$ Hydro reserve sold to market for regulation down provided by hydro generator j in hour t, when in generation mode [MWh]

$r_{jt}^{SR,G}$	Hydro reserve for spinning reserve provided by hydro generator j in hour t, when in generation mode [MWh]
$r_{t}^{RU,G}$	Hydro reserve sold to market for regulation up provided by all hydro generators in hour t [MWh]
$r_{t}^{RD,G}$	Hydro reserve sold to market for regulation down provided by all hydro generators in hour t [MWh]
$r_{t}^{SR,G}$	Hydro reserve for spinning reserve provided by all hydro generators in hour t, when in generation mode [MWh]
$y_{jt}^{G\ell+}$	Slack variables (upper bound) which take non-zero value if operation outside rough zone ℓ of hydro generator j cannot be honored in hour t, when operating in generation mode $[\mathrm{ft}^3/s]$
$y_{jt}^{G\ell-}$	Slack variables (lower bound) which take non-zero value if operation outside rough zone ℓ of hydro generator j cannot be honored in hour t, when operating in generation mode $[\mathrm{ft}^3/s]$

3. Problem Formulation

3.1 Objective Function

The forecast profit of hybrid Hydro-BESS firm during week w is given by (1):

$$\Omega_w = Max\left\{\sum_{t=1}^{N_H}\left(\Omega_t^E + \Omega_t^{AS} + \Omega_t^{RD} - \Phi_t^H\right)\right\} \tag{1}$$

Where:

Ω_t^E Revenue from the sale of energy in the day-ahead energy market [$]

Ω_t^{AS} Revenue from sale of ancillary services in the day-ahead market clearing, including regulation-up, regulation-down and spinning reserve [$]

Ω_t^{RD} Adjustments to energy revenue from deployment of regulation up/down [$]

Φ_t^H Operational expenses of hydroelectric plant, formed by variable operation and maintenance costs, and startup and shutdown costs plus penalties incurred if it is not possible to avoid operating within rough zone [$]

The above terms are given as follows:

$$\Omega_t^E = \lambda_t^E \cdot [p_t^{GGr} + p_t^{BD} + p_t^{RGr} - (1 + \epsilon) \cdot p_t^{GrSt}] \tag{2 - MOD}$$

$$\Omega_t^{AS} = \lambda_t^{RU} \cdot r_t^{RU} + \lambda_t^{RD} \cdot r_t^{RD} + \lambda_t^{SR} \cdot r_t^{SR} \tag{3 - MOD}$$

$$\Omega_t^{RD} = \lambda_t^{E,RT} \cdot [\delta_t^{RU} \cdot r_t^{RU} - \delta_t^{RD} \cdot r_t^{RD}] \tag{4 - MOD}$$

$$\Phi_t^H = \sum_{j\in\mathcal{U}}\left(\lambda_j^{O\&M} \cdot (p_{jt}^G + \delta_t^{RU} \cdot r_t^{RU,G} - \delta_t^{RD} \cdot r_t^{RD,G}) + SU_j^G \cdot a_{jt}^G + SD_j^G \cdot z_{jt}^G + SU_j^P \cdot a_{jt}^P + SD_j^P \cdot z_{jt}^P + \sum_{\ell\in Z_j} C^{rz}\left[y_{jt}^{G\ell+} + y_{jt}^{G\ell-}\right]\right) \tag{5 - MOD}$$

In publications such as [3], a uniform price is set for regulation services. However, the proposed objective function aligns with the one used in the CHEERS model, as well as in [4] where distinct prices for regulation-up and regulation-down are considered. The term (3) varies, depending on market, as exemplified in [1,4].

3.2 Constraints

3.2.1 Battery Energy Storage System (BESS)

(a) BESS energy balance equation, considering losses and regulation signal (deployment of up and down regulation):

$$e_t^B = e_{t-1}^B + \eta^C \cdot \left[p_t^{BC} + \delta_t^{RD} \cdot r_t^{RD,BC} - \delta_t^{RU} \cdot r_t^{RU,BC}\right] - \frac{1}{\eta^D} \cdot \left[p_t^{BD} + \delta_t^{RU} \cdot r_t^{RU,BD} - \delta_t^{RD} \cdot r_t^{RD,BD}\right] \quad \forall t \tag{6}$$

Notice that the balance equation includes the predicted deployment of regulation energy in real time, affected by losses.

(b) Limits on storage device state of charge, considering the provision of ancillary services:

$$e_{it}^B \geq SOC_i^- + \frac{1}{\eta_i^D}\left(r_{it}^{RU,BD} + r_{it}^{SR,BD}\right) \quad \forall i \in S, t = 1, \dots, N_H \qquad (7-MOD)$$

$$e_{it}^B \leq SOC_i^+ - \eta_i^C \cdot r_{it}^{RD,BC} \quad \forall i \in S, t = 1, \dots, N_H \qquad (8-MOD)$$

(c) Operational limits in discharging mode:

$$p_{it}^{BD} + r_{it}^{RU,BD} + r_{it}^{SR,BD} \leq P_i^{BD+} \cdot u_{it}^B \quad \forall i \in S, t = 1, \dots, N_H \tag{9}$$

$$p_{it}^{BD} - r_{it}^{RD,BD} \geq 0 \quad \forall i \in S, t = 1, \dots, N_H \tag{10}$$

$$0 \leq r_{it}^{RU,BD} \leq RU_i^{B+} \quad \forall i \in S, t = 1, \dots, N_H \tag{11}$$

$$0 \leq r_{it}^{RD,BD} \leq RD_i^{B+} \quad \forall i \in S, t = 1, \dots, N_H \tag{12}$$

$$0 \leq r_{it}^{SR,BD} \leq SR_i^{B+} \quad \forall i \in S, t = 1, \dots, N_H \tag{13}$$

(d) Operational limits in charging mode:

$$p_{it}^{BC} + r_{it}^{RD,BC} \leq P_i^{BC+} \cdot (1 - u_{it}^B) \quad \forall i \in S, t = 1, \dots, N_H \tag{14}$$

$$p_{it}^{BC} - r_{it}^{RU,BC} - r_{it}^{SR,BC} \geq 0 \quad \forall i \in S, t = 1, \dots, N_H \tag{15}$$

$$0 \leq r_{it}^{RU,BC} \leq RU_i^{B+} \quad \forall i \in S, t = 1, \dots, N_H \tag{16}$$

$$0 \leq r_{it}^{RD,BC} \leq RD_i^{B+} \quad \forall i \in S, t = 1, \dots, N_H \tag{17}$$

$$0 \leq r_{it}^{SR,BC} \leq SR_i^{B+} \quad \forall i \in S, t = 1, \dots, N_H \tag{18}$$

(e) Discharging and charging power of BESS:

$$p_t^{BD} = \sum_{i \in S} p_{it}^{BD} \quad \forall t \qquad (19-NEW)$$

$$p_t^{BC} = \sum_{i \in S} p_{it}^{BC} \quad \forall t \qquad (20-NEW)$$

(f) Initial storage of BESS:

$$e_0^B = SOC^{ini} \tag{21}$$

$$e_{N_H}^B = SOC^{ini} \tag{22 - NEW}$$

(g) Allocation of ancillary services provided by BESS units:

$$r_{it}^{RU,B} = r_{it}^{RU,BC} + r_{it}^{RU,BD} \qquad \forall i \in S, t = 1, \dots, N_H \tag{23 - NEW}$$

$$r_{it}^{RD,B} = r_{it}^{RD,BC} + r_{it}^{RD,BD} \qquad \forall i \in S, t = 1, \dots, N_H \tag{24 - NEW}$$

$$r_{it}^{SR,B} = r_{it}^{SR,BC} + r_{it}^{SR,BD} \qquad \forall i \in S, t = 1, \dots, N_H \tag{25 - NEW}$$

$$r_t^{RU,B} = \sum_{i \in S} r_{it}^{RU,B} \qquad \forall t \tag{26 - NEW}$$

$$r_t^{RD,B} = \sum_{i \in S} r_{it}^{RD,B} \qquad \forall t \tag{27 - NEW}$$

$$r_t^{SR,B} = \sum_{i \in S} r_{it}^{SR,B} \qquad \forall t \tag{28 - NEW}$$

(h) BESS annual energy throughput limit:

$$\sum_{t=1}^{N_H} \frac{1}{\eta_i^D} \cdot \left(p_{it}^{BD} + \delta_t^{RU} \cdot r_{it}^{RU,B}\right) \le \frac{N_D}{365} \cdot AET_i \qquad \forall i \in S, t = 1, \dots, N_H \tag{29A - NEW}$$

$$AET_i = \frac{\left(SOC_i^+ - SOC_i^-\right) \cdot L_i^{cyc}}{L_i^{cal}} \qquad \forall i \in S \tag{29B - NEW}$$

3.2.2 Pumped Storage Hydro (PSH) System

(i) Commitment of hydro generators and pumps:

$$u_{jt}^G - u_{j(t-1)}^G = a_{jt}^G - z_{jt}^G \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_H \tag{30}$$

$$a_{jt}^G + z_{jt}^G \le 1 \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_H \tag{31}$$

$$u_{jt}^G, a_{jt}^G \in \{0,1\} \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_H \tag{32}$$

$$0 \le z_{jt}^G \le 1 \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_H \tag{33}$$

$$u_{jt}^{P} - u_{j(t-1)}^{P} = a_{jt}^{P} - z_{jt}^{P} \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_{\mathrm{H}} \qquad (34 - NEW)$$

$$a_{jt}^{P} + z_{jt}^{P} \leq 1 \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_{\mathrm{H}} \qquad (35 - NEW)$$

$$u_{jt}^{P}, a_{jt}^{P} \in \{0,1\} \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_{\mathrm{H}} \qquad (36 - NEW)$$

$$0 \leq z_{jt}^{P} \leq 1 \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_{\mathrm{H}} \qquad (37 - NEW)$$

$$\sum_{j \in \mathcal{U}} u_{jt}^{G} \leq (1 - y_{t}^{P}) \cdot N_{G} \qquad \forall t \qquad (38A - NEW)$$

$$\sum_{j \in \mathcal{U}} u_{jt}^{P} \leq y_{t}^{P} \cdot N_{P} \qquad \forall t \qquad (38B - NEW)$$

(j) Operational limits of hydro generators considering generation and ancillary services, when in generation mode:

$$p_{jt}^{G} + r_{jt}^{RU,G} + r_{jt}^{SR,G} \leq P_{j}^{G+} \cdot u_{jt}^{G} \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_{\mathrm{H}} \qquad (39 - NEW)$$

$$p_{jt}^{G} - r_{jt}^{RD,G} \geq P_{j}^{G-} \cdot u_{jt}^{G} \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_{\mathrm{H}} \qquad (40 - NEW)$$

$$0 \leq r_{jt}^{RU,G} \leq RU_{j}^{G+} \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_{\mathrm{H}} \qquad (41 - NEW)$$

$$0 \leq r_{jt}^{RD,G} \leq RD_{j}^{G+} \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_{\mathrm{H}} \qquad (42 - NEW)$$

$$0 \leq r_{jt}^{SR,G} \leq SR_{j}^{G+} \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_{\mathrm{H}} \qquad (43 - NEW)$$

(k) Ramping constraints when in generation mode:

$$\left(p_{jt}^{G} + r_{jt}^{RU,G}\right) - \left(p_{j(t-1)}^{G} - r_{j(t-1)}^{RD,G}\right) \leq R_{j}^{U,G} \cdot u_{j(t-1)}^{G} + R_{j}^{SU,G} \cdot a_{jt}^{G} \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_{\mathrm{H}} \qquad (44 - MOD)$$

$$\left(p_{j(t-1)}^{G} + r_{j(t-1)}^{RU,G}\right) - \left(p_{jt}^{G} - r_{jt}^{RD,G}\right) \leq R_{j}^{D,G} \cdot u_{jt}^{G} + R_{j}^{SD,G} \cdot z_{jt}^{G} \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_{\mathrm{H}} \qquad (45 - MOD)$$

(l) Water discharge of plant:

$$q_{jt}^{G1} \leq Q_{j}^{G1+} \cdot u_{jt}^{G} \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_{H} \qquad (46)$$

$$q_{jt}^{G1} \geq Q_{j}^{G1+} \cdot w_{jt}^{G1} \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_{\mathrm{H}} \qquad (47)$$

$$q_{jt}^{G\ell} \leq Q_j^{G\ell+} \cdot w_{jt}^{G\ell-1} \qquad \forall l \geq 2 \in \mathcal{L}_j, \forall j \in \mathcal{U}, t = 1, \dots, N_{\mathrm{H}} \tag{48}$$

$$q_{jt}^{G\ell} \geq Q_j^{G\ell+} \cdot w_{jt}^{G\ell} \qquad \forall l \geq 2 \in \mathcal{L}_j, \forall j \in \mathcal{U}, t = 1, \dots, N_H \tag{49}$$

$$q_{jt}^{G} = \sum_{\ell \in \mathcal{L}_j} q_{jt}^{G\ell} + Q_j^{G-} \cdot u_{jt}^{G} \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_{\mathrm{H}} \tag{50}$$

(m) Flow equivalent of ancillary services provided by hydro generators, when in generation mode:

$$q_{jt}^{G} + q_{jt}^{RU,G} + q_{jt}^{SR,G} \leq \left(Q_j^{G-} + \sum_{\ell \in \mathcal{L}_j} Q_j^{G\ell+} \right) \cdot u_{jt}^{G} \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_H \tag{51 – NEW}$$

$$q_{jt}^{G} - q_{jt}^{RD,G} \geq Q_j^{G-} \cdot u_{jt}^{G} \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_H \tag{52 – NEW}$$

(n) Generation of hydro power and allocation of ancillary services, when in generation mode:

$$p_{jt}^{G} = P_j^{G-} \cdot u_{jt}^{G} + \sum_{\ell \in \mathcal{L}_j} \rho_j^{G\ell} \cdot q_{jt}^{G\ell} \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_{\mathrm{H}} \tag{53}$$

$$p_t^{G} = \sum_{j \in \mathcal{U}} p_{jt}^{G} \qquad \forall t \tag{54}$$

$$r_{jt}^{RU,G} = \hat{\rho}_j^{G} \cdot q_{jt}^{RU,G} \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_H \tag{55 – NEW}$$

$$r_{jt}^{RD,G} = \hat{\rho}_j^{G} \cdot q_{jt}^{RD,G} \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_H \tag{56 – NEW}$$

$$r_{jt}^{SR,G} = \hat{\rho}_j^{G} \cdot q_{jt}^{SR,G} \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_H \tag{57 – NEW}$$

$$r_t^{RU,G} = \sum_{j \in \mathcal{U}} r_{jt}^{RU,G} \qquad \forall t \tag{58 – NEW}$$

$$r_t^{RD,G} = \sum_{j \in \mathcal{U}} r_{jt}^{RD,G} \qquad \forall t \tag{59 – NEW}$$

$$r_t^{SR,G} = \sum_{j \in \mathcal{U}} r_{jt}^{SR,G} \qquad \forall t \tag{60 – NEW}$$

(o) Consumption of hydro power, when in pumping mode:

$$p_{jt}^{P} = \rho_j^{P} \cdot q_{jt}^{P} + u_{jt}^{P} \cdot P_j^{P-} \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_{\mathrm{H}} \tag{61 – NEW}$$

$$p_t^P = \sum_{j \in \mathcal{U}} p_{jt}^P \qquad \forall t \qquad (62 - NEW)$$

$$q_{jt}^P + \frac{1}{\rho_j^P} \cdot P_j^{P-} \cdot u_{jt}^P \leq u_{jt}^P \cdot Q_j^{P+} \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_\mathrm{H} \qquad (63 - NEW)$$

$$q_{jt}^{P-TOT} = q_{jt}^P + \frac{1}{\rho_j^P} \cdot P_j^{P-} \cdot u_{jt}^P \qquad \forall j \in \mathcal{U}, t = 1, \dots, N_\mathrm{H} \qquad (64 - NEW)$$

(p) Water balance equation:

$$e_t^H = e_{t-1}^H + CF \cdot \left(\sum_{j \in \mathcal{U}} \left[q_{jt}^P + \frac{1}{\rho_j^P} \cdot P_j^{P-} \cdot u_{jt}^P - q_{jt}^G - \delta_t^{RU} \cdot q_{jt}^{RU,G} + \delta_t^{RD} \cdot q_{jt}^{RD,G} \right] - C_0^{filt} - C_1^{filt} \cdot e_t^H \right) \quad \forall t \qquad (65 - MOD)$$

(q) Volume limits:

$$e_t^H \geq V^- + CF \cdot \sum_{j \in \mathcal{U}} \left(q_{jt}^{RU,G} + q_{jt}^{SR,G} \right) \qquad \forall t \qquad (66 - MOD)$$

$$e_t^H \leq V^+ - CF \cdot \sum_{j \in \mathcal{U}} \left(q_{jt}^{RD,G} \right) \qquad \forall t \qquad (67 - MOD)$$

(r) Initial and End-of-Period water volume constraints:

$$e_0^H = V^- + (V^+ - V^-) \cdot (1 - \varphi^{ini}) \qquad (68 - MOD)$$

$$e_{N_H}^H \geq V^- + (V^+ - V^-) \cdot (1 - \varphi^{end}) \qquad (69 - MOD)$$

(s) Rough zone constraints:

$$\left(q_{jt}^{G\ell} - y_{jt}^{G\ell-} \right) \geq 0 \qquad \forall j \in \mathcal{U}, \ell \in \mathcal{Z}_j, t = 1, \dots, N_H \qquad (70)$$

$$\left(Q_j^{G\ell+} - q_{jt}^{G\ell} - y_{jt}^{G\ell+} \right) \geq 0 \qquad \forall j \in \mathcal{U}, \ell \in \mathcal{Z}_j, t = 1, \dots, N_H \qquad (71)$$

$$\left(q_{jt}^{G\ell} - y_{jt}^{G\ell-} \right) \leq M \cdot \psi_{jt}^{G\ell} \qquad \forall j \in \mathcal{U}, \ell \in \mathcal{Z}_j, t = 1, \dots, N_H \qquad (72)$$

$$\left(Q_j^{G\ell+} - q_{jt}^{G\ell} - y_{jt}^{G\ell+} \right) \leq M \cdot \left(1 - \psi_{jt}^{G\ell} \right) \qquad \forall j \in \mathcal{U}, \ell \in \mathcal{Z}_j, t = 1, \dots, N_H \qquad (73)$$

$$\psi_{jt}^{G\ell} \in \{0,1\} \qquad \forall j \in \mathcal{U}, \ell \in \mathcal{Z}_j, t = 1, \dots, N_H \qquad (74)$$

$$y_{jt}^{G\ell+}, y_{jt}^{G\ell-} \geq 0 \qquad \forall j \in \mathcal{U}, \ell \in \mathcal{Z}_j, t = 1, \dots, N_H \tag{75}$$

$$y_{jt}^{G\ell+}, y_{jt}^{G\ell-} \leq Q_j^{G\ell+} \qquad \forall j \in \mathcal{U}, \ell \in \mathcal{Z}_j, t = 1, \dots, N_H \tag{76}$$

(t) Startups daily limit of hydro generators and hydro pumps:

$$\sum_{t=(d-1)\cdot 24+1}^{d\cdot 24} a_{jt}^{G} \leq SU_j^{G,day} \qquad \forall j \in \mathcal{U}, \forall d = 1, \dots, N_D \tag{77 – NEW}$$

$$\sum_{t=(d-1)\cdot 24+1}^{d\cdot 24} a_{jt}^{P} \leq SU_j^{P,day} \qquad \forall j \in \mathcal{U}, \forall d = 1, \dots, N_D \tag{78 – NEW}$$

3.2.3 RES generators constraints

(u) Non-dispatchable renewable generation constraints:

$$p_{kt}^{R} + p_{kt}^{RSpill} = P_{kt}^{RFor} \qquad \forall k \in \mathcal{R}, \forall t = 1, \dots, N_H \tag{79 – NEW}$$

$$p_t^{R} = \sum_{k \in \mathcal{R}} p_{kt}^{R} \qquad \forall t \tag{80 – NEW}$$

$$p_{kt}^{R}, p_{kt}^{RSpill} \geq 0 \qquad \forall k \in \mathcal{R}, \forall t = 1, \dots, N_H \tag{81 – NEW}$$

3.2.4 Coupling Constraints

(v) Ancillary services offered to the market:

(1) Reserve sold for Regulation Up:

$$r_t^{RU} = r_t^{RU,BD} + r_t^{RU,BC} + r_t^{RU,G} \tag{82 – MOD}$$

(2) Reserve sold for Regulation Down:

$$r_t^{RD} = r_t^{RD,BD} + r_t^{RD,BC} + r_t^{RD,G} \tag{83 – MOD}$$

(3) Reserve sold for Spinning Reserve:

$$r_t^{SR} = r_t^{SR,BD} + r_t^{SR,BC} + r_t^{SR,G} \tag{84 – MOD}$$

(w) Hydro + BESS + RES power balances and interconnection limits:

$$p_t^{G} = p_t^{GGr} + p_t^{GB} \qquad \forall t \tag{85}$$

$$p_t^R = p_t^{RGr} + p_t^{RSt} \quad \forall t \qquad (86-NEW)$$

$$p_t^{BC} + p_t^P = p_t^{GB} + p_t^{RSt} + p_t^{GrSt} \quad \forall t \qquad (87-MOD)$$

$$p_t^{BD} + p_t^{GGr} + p_t^{RGr} + r_t^{RU} + r_t^{SR} \le T^{exp+} \quad \forall t \qquad (88-MOD)$$

$$p_t^{GrSt} + r_t^{RD} \le T^{imp+} \quad \forall t \qquad (89-MOD)$$

4. Regulation Signal

Regulation signal can be found at:
https://www.pjm.com/markets-and-operations/ancillary-services

Scroll down to **Regulation Self-Test Signals**. Choose the link and download file "Normalized Dynamic and Traditional Regulation Signals - May 2014", which is and .xls file. In that file, choose the Dynamic (RegD) worksheet.

5. Consideration of rough zones

Constraints (43)-(47) establish the dynamic of water release according to the operational region of the turbine-generator set. These operational regions are exemplified in Table 1 for a hydro generator with a performance curve divided in 4 blocks, where for simplicity only the block number is indexed.

Table 1: Operating regions of hydro generator with 4 blocks in performance curve.

	Status of w^ℓ				Bounds on $q^{G\ell}$			
Block	w^1	w^2	w^3	w^4	q^{G1}	q^{G2}	q^{G3}	q^{G4}
1	0	0	0	0	$0 \le q^{G1} \le Q^{G1+}$	$q^{G2} = 0$	$q^{G3} = 0$	$q^{G4} = 0$
2	1	0	0	0	$q^{G1} = Q^{G1+}$	$0 \le q^{G2} \le Q^{G2+}$	$q^{G3} = 0$	$q^{G4} = 0$
3	1	1	0	0	$q^{G1} = Q^{G1+}$	$q^{G2} = Q^{G2+}$	$0 \le q^3 \le Q^{G3+}$	$q^{G4} = 0$
4	1	1	1	0	$q^{G1} = Q^{G1+}$	$q^{G2} = Q^{G2+}$	$q^{G3} = Q^{G3+}$	$0 \le q^{G4} \le Q^{G4+}$

Suppose that block 3 is identified as a rough zone for the hydro generator, so that generation within that range should be avoided. From Table 1, it is seen that this is possible by either keeping $q_3 = 0$ or $q_3 = U^{G3+}$. This gives rise to the following complementarity constraint:

$$q_3 \cdot (Q^{G3+} - q_3) = 0 \qquad (90)$$

which can also be expressed as:

$$q_3 \ge 0 \perp (Q^{G3+} - q_3) \ge 0 \qquad (91)$$

If honoring the rough zone constraint is infeasible, the following slack variables are used, so that a solution by the MILP solver is always possible:

$$(q_3 - y_3^{G-}) \geq 0 \perp (Q^{G3+} - q_3 - y_3^{G+}) \geq 0 \tag{92}$$

and y_3^{G+} and y_3^{G-} are penalized in the objective function, using a large scalar. After linearizing the above constraints with the "big M″ method, these conditions are replaced by:

$$\left(q_3 - y_3^{G-}\right) \geq 0 \tag{93}$$

$$(Q^{G3+} - q_3 - y_3^{G+}) \geq 0 \tag{94}$$

$$(q_3 - y_3^{G-}) \leq \psi_3 \cdot M \tag{95}$$

$$(Q^{G3+} - q_3 - y_3^{G+}) \leq (1 - \psi_3) \cdot M \tag{96}$$

$$0 \leq y_3^{G-} \leq Q^{G3+} \tag{97}$$

$$0 \leq y_3^{G+} \leq Q^{G3+} \tag{98}$$

$$\psi_3 \in \{0,1\} \tag{99}$$

where M is a large enough scalar.

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